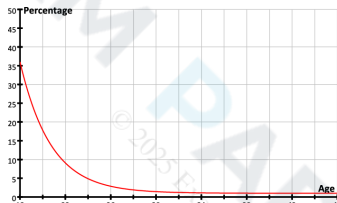


Question			Answer/Indicative content	Marks	Part marks and guidance	
1			With no more than 3 points on his licence the driver's premium is not altered: £520 The driver now has $3 + 6 = 9$ points. The new premium is $£520 \times 2^{\frac{9}{6}} = £1470.78$	B1 B1	£520 £1470.78 or £1471 [but not £1470.80] Examiner's Comments Most candidates were successful here. A few did not round a money answer to an appropriate accuracy.	
			Total	2		
2		i		B1	A sketch. Must be at least for values of x between 18 and 45. Must be correct shape ie descending curve, does not curve up at end, does not cross the horizontal axis (even if extended) ie must $\rightarrow 1$. Does not need to go through points exactly as a sketch. Examiner's Comments The graph was generally good. The commonest error was that it often stopped at $x=44$ instead of continuing to at least $x=45$. In other cases the graph looked as though it would cut through the x axis when produced.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	For large values of x , $be^{-k(x-17)} \rightarrow 0$, and so $y \rightarrow a$ which is 1 in this case.	B1	Need x becoming large or $\rightarrow \infty$, exponential term $\rightarrow 0$ and $y \rightarrow a$, $a = 1$ or $y = 1$, $a = 1$ oe NOT just at $x = 45$, $y = 1$ Examiner's Comments Few candidates were able to give a complete solution that as $x \rightarrow \infty$, $be^{-k(x-17)} \rightarrow 0$ and so as $y \rightarrow a$, $a = 1$ or equivalent. Many referred to the general shape of the curve or asymptotes or said that at $x=45$, $y=1$. Generally the responses were unclear or incomplete.	
		iii	Substituting $x = 23 \Rightarrow y = 6.6$	M1	subst $x = 23$ and finding $y = 6.6$ (6.60115...) (or subst $y = 6$ and finding $x = 23.3$ (23.3097...)) Only accept comparing x or y, not coefficients	
		iii	This is quite close to the observed value of $y = 6$.	A1	(or close to $x = 23$) Accept say, $y = 6.6 \approx 6$. But not say at $x = 23$, $y \approx 6$ with no evaluation seen Accept if states, say, $y = 6.6$ which is not consistent with $y = 6$ oe Examiner's Comments This was well answered although some candidates forgot to compare the result of their calculation with the given point. Most compared 6 with 6.60... but some used 23. A few tried to establish the given constants b and k.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance																			
3			<table><tr><th>Year</th><th>% discount</th></tr><tr><td>2007</td><td>0</td></tr><tr><td>2008</td><td>30</td></tr><tr><td>2009</td><td>40</td></tr><tr><td>2010</td><td>50</td></tr><tr><td>2011</td><td>60</td></tr><tr><td>2012</td><td>65</td></tr><tr><td>2013</td><td>65</td></tr><tr><td>Total</td><td>310</td></tr></table>	Year	% discount	2007	0	2008	30	2009	40	2010	50	2011	60	2012	65	2013	65	Total	310	B1	obtaining 310 or 3.1 oe [from adding all relevant terms ie 0 + 30 + 40 + 50 + 60 + 65 + 65 = 310 or 0 + 0.3 + 0.4 + 0.5 + 0.6 + 0.65 + 0.65 = 3.1 soi (with or without first zero term) or from 700 - 100 - 70 - 60 - 50 - 40 - 2 × 35 oe]	
			Year	% discount																				
			2007	0																				
			2008	30																				
			2009	40																				
			2010	50																				
			2011	60																				
			2012	65																				
			2013	65																				
			Total	310																				
310% of the premium is £3875	M1	3875 ÷ their 3.1 even if one term was missing from addition but must come from attempt at the appropriate addition [ie an error in adding to 310 or an omission of one term, an inclusion of say an extra 65, or an addition of 100, 70, 60,...etc with subsequent subtraction from 700]																						
The premium is																								
$\frac{£3875 \times 100}{310}$																								
	A1	£1250 cao																						
		Examiner's Comments																						
		This was a good discriminator on Paper B. There were many good solutions scoring full marks. Others often included the wrong number of 0.65s in their total or sometimes incorrectly used 0.7+0.6+0.5+0.4.....and failed to subtract. Some others had no idea and tried to find say 65% of £3875.																						

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	3		
4			Basic premium = 35% of driver's premium Drivers premium = Basic premium $\div$ 0.35 = 2.86 Basic premium $\Rightarrow k = 2.86$	M1 A1	use of 35% or 0.35 oe [not 65, 0.65 unless 1-0.65] accept 2.86 or better ($k = 2.85714\dots$ or $\frac{26}{7}$ oe) [$k = 2.9$, $k = 1/0.35$ scores M1A0] Examiner's Comments This was not particularly well answered. Candidates seemed almost equally to find correctly that $k=1/0.35=2.86$ or to use $1/0.65$. A surprising number used 1-0.65 as 0.45.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance																					
5		i	<table><tr><td>Group</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>Number of people</td><td>15</td><td>30</td><td>60</td><td>45</td></tr><tr><td>Average number of accidents in a year</td><td>1.5</td><td>4.5</td><td>3</td><td>9</td></tr><tr><td>Average cost of accidents per year</td><td>£7500</td><td>£9000</td><td>£3000</td><td>£4500</td></tr></table>	Group	P	Q	R	S	Number of people	15	30	60	45	Average number of accidents in a year	1.5	4.5	3	9	Average cost of accidents per year	£7500	£9000	£3000	£4500	B1	for all three entries 30, 60, 45 correct	
	Group	P	Q	R	S																					
	Number of people	15	30	60	45																					
	Average number of accidents in a year	1.5	4.5	3	9																					
Average cost of accidents per year	£7500	£9000	£3000	£4500																						
	i		B1	for all three entries 1.5, 3, 9 correct																						
	i		B1	for 3000, 4500 both correct																						
	i			SC B2 for all entries in any three columns correct Examiner's Comments Most candidates correctly completed the table.																						
		ii	£24000 ÷ 150	M1	Adding their bottom row (7500 + 9000 + '3000' + '4500' = '24000') and dividing by 150 soi (and not divided or multiplied by any additional values)																					
		ii	=£160	A1	ft their 24000÷150 (corr to 2dp if inexact) Examiner's Comments £24000/150=£160 was usually correct. Some candidates multiplied or divided by additional values, usually 18 or 9.																					
		iii	The 45 members of Group S pay 1.5 × £4500 = £6750. So each pays £6750÷45	M1	their 4500 × 1.5 oe soi as part of solution																					

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	=£150.	A1	cao www (must be from correct final column) Examiner's Comments This was often correct but often long methods were seen including $£24000 \times 1.5 = £36000$, then $18.75\% \text{ of } £36000 = £6750$ instead of the more direct approach of $£4500 \times 1.5 = £6750$. Some candidates stopped at £6750.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6			$n = 15, f = 6$ gives 5.6	B1	5.6	
			$n = 15, f = 4$ gives 3.9	B1	3.9	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			so 'usually' 6 and 4 stops respectively	E1	Dependent on B1B1 – additional values of f considered E0 Examiner's Comments Part (i) was answered extremely well with nearly all candidates correctly expressing $\frac{1}{(1+2x)(1-x)}$ in partial fractions. In part (ii) the majority of candidates were able to separate the variables and substitute their partial fractions correctly. There were, however, frequent errors in the integration usually when candidates forgot to divide by 2 when integrating $\frac{1}{1+2x}$ and/or when they forgot to do the same process with the $-\frac{1}{1-x}$. Many candidates did not include a constant of integration or, if included, it was either subsequently ignored or set to zero without any mathematical justification. Most candidates were able to combine their logarithmic terms correctly, though examiners noted the high volume of cases in which the 'correct' printed answer was seen following earlier incorrect working. Most candidates achieved the first two marks in part (iii) for finding the value of k, but many made errors in	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					handling the logarithms to find the time taken for the drug concentration to reach 90% of its maximum value. Examiners noted the large variation in the accuracy of candidates' final answers in this part. In part (iv) most candidates multiplied up by $1-x$, collected and factorised the x terms correctly. The main problem seemed to be how to get the negative exponent. These often appeared when candidates divided $e^{3kt} - 1$ by $2 + e^{kt}$, losing the final accuracy mark in the process. It was also common for candidates to simply not show how $\frac{e^{3kt} - 1}{2 + e^{kt}}$ was equal to the given answer. Those candidates who started by taking the reciprocal of the answer given in part (ii) were usually far more successful in deriving the required result in this part. The majority of candidates who attempted to verify that the drug concentration approached its maximum value in the long term recognised that as $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$, although some candidates simply substituted a large value of t to show that x was close to 1, this approach was not sufficient to earn the final mark in this part.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7			$1 - \left(\frac{9}{10}\right)^8 = 0.5695\dots$	M1	Substitution of the correct values into the correct formula	
				A1	Accept 2 to 4 dp	
				Examiner's Comments		
				There seemed to be a misconception amongst a number of candidates about the definition of 'show' and 'verify' in this first part. The majority of candidates correctly substituted the coordinates of A, B and E into the equation of the plane, to achieve the first two marks. Those candidates who tried to derive the equation of the plane had methods that were generally incomplete and despite a lot of work, did not gain much credit.		
				In part (ii)(A) a number of candidates seemed to think that just showing one direction vector was normal to the plane was sufficient and some candidates showed all three. It was unfortunate that so many lost marks by not showing the evaluation of the scalar product(s) even though this was another 'show that' question and so examiners had to be convinced that the candidates were indeed showing the required results and not simply stating them.		
				Part (ii)(B) was nearly always done correctly with only a small minority having the equation of the plane as $x - 6y - 6 = 0$ rather than the		

Question	Answer/Indicative content	Marks	Part marks and guidance
			correct $x - 6y + 6 = 0$. Part (ii)(C) occasionally contained simple arithmetical errors with the correct equation often leading to either $b = b = \frac{3}{4}$ or $b = -b = -\frac{4}{3}$. Some candidates stated the direction vector $\overrightarrow{FG}$ correctly and then either did not state the length of FG or resorted to using Pythagoras to calculate this length, not recognising that they could simply read off the required answer. To find the angle EFB in part (iii) most candidates attempted to find vectors (and hence use the scalar product) that would lead to the correct angle, though there were common arithmetical errors here. Some candidates used $\overrightarrow{EF} \cdot \overrightarrow{FB}$ and hence obtained an acute angle, often not recognising that they needed to subtract this from 180 as angle EFB was obtuse. Some candidates who used $\overrightarrow{FE}$ and $\overrightarrow{FB}$ 'lost' the minus sign in their calculation and ended up with an answer of 36.9°. A very small number of candidates calculated the lengths of the three sides of triangle EFB and used the cosine rule. Part (iv) required

Question			Answer/Indicative content	Marks	Part marks and guidance	
					candidates to find the height of Q above the ground. Those candidates who thought the y – axis represented the vertical height clearly did not look at the diagram carefully enough. A surprising number of candidates could not calculate the z coordinate of P correctly. Some candidates seemed to think that in finding $\overline{PQ}$ they had arrived at the required answer whereas other candidates worked out the direct distance from the origin to point Q. Too many candidates worked out both the x and y coordinates of Q when only the z coordinate was required. Finally, a number of candidates found the correct position vector for Q but did not explicitly state the height of Q above the ground.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance																									
8		i	<table><tr><td></td><td colspan="2">Car C</td></tr><tr><td></td><td>Arrival time (seconds)</td><td>Departure time (seconds)</td></tr><tr><td>Ground floor</td><td>0</td><td>20</td></tr><tr><td></td><td></td><td></td></tr><tr><td>Floor 7</td><td>41</td><td>56</td></tr><tr><td>Floor 8</td><td>59</td><td>74</td></tr><tr><td></td><td></td><td></td></tr><tr><td>Return to ground floor</td><td>98</td><td></td></tr></table>		Car C			Arrival time (seconds)	Departure time (seconds)	Ground floor	0	20				Floor 7	41	56	Floor 8	59	74				Return to ground floor	98		B1		
	Car C																													
	Arrival time (seconds)	Departure time (seconds)																												
Ground floor	0	20																												
Floor 7	41	56																												
Floor 8	59	74																												
Return to ground floor	98																													
		i		B1	59, 74 or their 56 + 3, their 59 + 15																									
		i		B1	98 or their 74 + 24																									

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	4×98s = 392s (6 mins 32s)	B1	cao Examiner's Comments In part (i) the majority of candidates correctly wrote down the result that $\frac{dy}{dx} = \frac{2\sec^2 \theta}{\sec \theta \tan \theta}$ It was clear though that a significant number of candidates did not know that $\frac{d}{d\theta}(\tan \theta) = \sec^2 \theta$, and the majority of these applied the quotient rule to $\frac{\sin \theta}{\cos \theta}$ in an attempt to derive this result. A significant minority of candidates did not show adequate working to gain the final mark in part (i) with many going directly from $\frac{2\sec \theta}{\tan \theta}$ (or equivalent) to the given answer. Candidates are reminded that in questions in which the demand is to show a given result then sufficient working must be shown to satisfy the examiner that the answer has been correctly derived and not simply written down. It was also noted by examiners that a number of candidates multiplied their expressions	

Question			Answer/Indicative content	Marks	Part marks and guidance
					for $\frac{dy}{d\theta}$ and $\frac{dx}{d\theta}$ rather than dividing. Although the techniques of verifying the Cartesian equation of the curve was relatively straightforward, answers in part (ii) were poorly presented, with many candidates unable to present a mathematical justification clearly and formally. Some candidates were insufficiently confident in the use of standard trigonometric identities and for some candidates it was as if they had never seen the relationship that $1 + \tan^2 \theta \equiv \sec^2 \theta$. In part (iii) the vast majority of candidates considered both the correct integral for the volume of revolution and integrated $4x^2 - 4$ correctly. It was the mention of a rotation of 180 that seemed to concern many candidates and a considerable number divided the correct answer of $1 + \tan^2 \theta \equiv \sec^2 \theta$ by 2.
			Total	4	

Question			Answer/Indicative content	Marks	Part marks and guidance	
9			Lift calls at floors 1 to 8 (only), leaving floor 8 at 164 seconds. Descent takes 24 seconds	B1*	Correct calculation eg $164 + 24$ or $230 - (30 + 12)$ (oe) but not $200 - 12$	
				E1dep*	Correct justification of the correct calculation (dependent on B1). If the departure time of 164 is used then their explanation must imply the use of the first 8 floors (not just '8 floors') Examiner's Comments This question differentiated well due to the coefficient of $\sin x$ taking the form of a positive constant rather than a number. Many candidates, however, were unfazed by this and worked out the correct values for R and α. Some candidates lost the first method mark by not including R in the expanded trigonometric statements $R\cos \alpha = 1$, $R\sin \alpha = \lambda$. Writing α in terms of the more complex $\arcsin$ and $\arccos$ expressions was surprisingly common. It was a little worrying that a sizeable minority of candidates went from the correct $R = \sqrt{1 + \lambda^2}$ to the incorrect $R = 1 + \lambda$, thinking the squared terms and the square root cancelled each other out. In part (ii) those candidates that realised that $R=2$ usually went on to get the correct values for λ and α. However it was common for λ to be incorrect due to an	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					incorrect expression for R from part (i). A fair proportion of candidates gave α in degrees and those who gave α as either $\arccos\left(\frac{1}{\sqrt{1+\lambda^2}}\right) \text{ or } \arcsin\left(\frac{\lambda}{\sqrt{1+\lambda^2}}\right)$ were generally less successful in this part than those who gave α as $\arctan \lambda$.	
			Total	2		

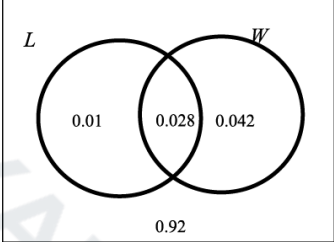
Question			Answer/Indicative content	Marks	Part marks and guidance	
10			$10\left(1 - \left(\frac{9}{10}\right)^7\right) = 5.22$	M1	Substitution of the correct values into the correct formula	
				A1	5.22 (must be 2 dp), B2 5.22 www (isw after correct answer seen) Examiner's Comments The most common mistake in part (i) was to use a value of 2 rather than -2 as the coefficient of x in each term of the expansion. The binomial coefficients were nearly always correct though a small number missed the 2! from the denominator of the x^2 term. While the majority of candidates used the correct value of n a small minority incorrectly used $n = n = \frac{1}{3}$ or $-\frac{2}{3}$. The range of validity of the expansion was done much better than in previous years although the most common mistake was to give non-strict inequalities. Other mistakes included: $-\frac{1}{2} < -x < \frac{1}{2}$ $ x > -\frac{1}{2}$ $\frac{1}{2} < x < -\frac{1}{2}$ In part (ii) the majority of candidates correctly multiplied their answer from part (i) with $(1-3x)$ and simplified this expression correctly to obtain the	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					correct values of a and b . It was concerning, however, that a number of candidates $(1-3x)(1+\frac{2}{3}x+\frac{8}{9}x^2+\dots)=1+\frac{2}{3}x+\frac{8}{9}x^2-3x-2x-\frac{8}{3}x^2+$ wrote or even more worryingly expanded $(1-3x)^1$ as $1-3x + \text{higher order terms in } x$.	
			Total	2		

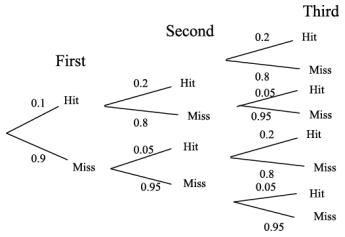
Question			Answer/Indicative content	Marks	Part marks and guidance
11			$\frac{110}{146} \times 100\% = 75.3\%$	M1	M1 allow (90 or 95 or 125) / 146×100
				A1	75.3 or better (eg 75.34 etc), 75 is A0 B2 75.3 with no working, B1 0.753 (oe) Examiner's Comments The majority of candidates correctly replaced $\cos 2\theta$ with $1 - 2 \sin^2 \theta$ although a minority of candidates made the costly mistake of replacing $\cos 2\theta$ with $1 - \sin^2 \theta$. While some candidates struggled to factorise $12 \sin^2 \theta - \sin \theta - 6 = 0$ many used the quadratic formula to solve this equation, and as with question 1, there were some candidates who did not state or apply the quadratic formula correctly. While the majority of candidates found the correct values for $\sin \theta$ some incorrectly obtained – $-\frac{3}{4}$ and/or $\frac{2}{3}$. Of those candidates that obtained the correct values for $\sin \theta$ the majority went on to score full marks. However, it was fairly common to see 'arcsin $\left(-\frac{2}{3}\right) = -41.81\ldots$ therefore no solutions in the range' with

Question			Answer/Indicative content	Marks	Part marks and guidance	
					no appreciation that solutions in the correct range could be found from this value. Having found the principal values it was common for candidates to get the other solutions in the range, often sketching the sine curve to help them, though most did this correctly without demonstrating any method.	
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	E2. Pick up Q	B1		
		ii	E5. Drop off Q	B1		
		iii	E4. Drop off P and R	B1	Examiner's Comments Common errors included: • $5(x+1)-3(2x+1)$ (and not the correct $5x(x+1)-\dots$) • Expanding $-3(2x+1)$ as either $-6x+3$ or $-6x+1$ • There were some candidates who did not multiply up on the right-hand side and so obtained $5x(x+1)-3(2x+1)=1$ • Some lost the final two marks for not applying the quadratic formula correctly However, this question was generally done well with most candidates scoring full marks and demonstrating sound basic algebraic manipulation skills. It was common to see the use of the quadratic formula as much as factorising to solve the final quadratic equation. Very few completed the square but those that did were mainly successful.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$P(L \cap W) = P(L W) \times P(W)$ $= 0.4 \times 0.07 = 0.028$	M1	For $P(L W) \times P(W)$	
		i		A1	cao Examiner's Comments This question was well answered, with about 80% of candidates scoring both marks.	
		ii		B1	For two labelled intersecting circles	
		ii		B1	For at least 2 correct probabilities.	FT their 0.028 provided < 0.038
		ii		B1	For remaining probabilities Examiner's Comments Almost all candidates gained the first mark for two labelled intersecting circles. Many candidates put their answer from part (i) into the intersection but then did not subtract their value from $P(W)$ so put 0.07 instead of $(0.07 - \text{their answer to (i)})$ in the other part of the circle labelled W. However, a reasonable number of candidates gained full credit, either having the correct 0.01 and 0.92 in the other parts or by following through correctly.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$P(L \cap W) = 0.028$, $P(L) \times P(W) = 0.038 \times 0.07 = 0.00266$	M1	For correct use of $P(L) \times P(W)$ If $P(L)$ wrong, max M1A0E0. No marks if $P(W)$ wrong	Or EG $P(L W) = 0.4$, $P(L) = 0.038$ Not equal so not independent M1 is for comparing with some attempt at numbers $P(L W)$ with $P(L)$, A1 for 0.038 If $P(L)$ wrong, max M1A0E0
		iii		A1	For 0.00266	
		iii	Not equal so not independent	E1* dep on	Allow 'they are dependent' Do not award E1 if	
		iii		M1	$P(L \cap W)$ wrong Examiner's Comments The vast majority of candidates tried to show non-independence by comparing $P(L) \times P(W)$ with $P(L \text{ intersect } W)$. However most of these did not have the correct value of $P(L)$ and many had $P(W)$ wrong, despite its value being given in the question. A small number of candidates compared $P(L W)$ with $P(L)$ and these were more often successful.	
			Total	8		

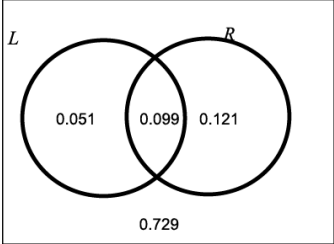
Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i		G1	For first set of branches	All probabilities correct
		i		G1	For second set of branches (indep)	All probabilities correct
		i		G1	For third set of branches (indep)	All probabilities correct
		i		G1	For labels <u>Examiner's Comments</u> The majority of tree diagrams were well constructed with correct labelling. Weaker candidates sometimes became confused and made errors in the 2nd and/or 3rd branch.	All correct labels for 'Hit' and 'Miss', 'H' and 'M' etc. Condone omission of First, Second, Third. Do not allow misreads here as all FT
		ii	P(Hits with at least one) = 1 – P(misses with all)	M1*	For $0.9 \times 0.95 \times 0.95$	FT their tree for both M marks, provided three terms
		ii	$= 1 - (0.9 \times 0.95 \times 0.95) = 1 - 0.81225 = 0.18775$	M1*dep	For 1 – ans	
		ii		A1	CAO	0.188 or better. Condone 0.1877 Allow 751/4000
		ii	ALTERNATIVE METHOD only if there is an attempt to add 7 probabilities			
		ii	At least three correct triple products	M1		
		ii	Attempt to add 7 triple products	M1		(not necessarily correct triple products)
		ii		A1	CAO	
		ii	FURTHER ALTERNATIVE METHOD			
		ii	$0.1 + 0.9 \times 0.05$	M1		
		ii	Above probability + $0.9 \times 0.95 \times 0.05$	M1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	CAO <u>Examiner's Comments</u> Many candidates employed the $1 - P(\text{misses with all})$ method, usually successfully, but a significant number used the protracted method of listing all 7 triplets associated with at least one hit. Usually errors were made using such an approach.	
		ii	P(Hits with exactly one)	M1	For two correct products	FT their tree for all three M marks, provided three terms Allow 0.155 or better
		ii	$= (0.1 \times 0.8 \times 0.95) + (0.9 \times 0.05 \times 0.8) + (0.9 \times 0.95 \times 0.05)$	M1	For all three correct products	
		ii	$= 0.076 + 0.036 + 0.04275 = \frac{19}{250} + \frac{9}{250} + \frac{171}{4000}$	M1	For sum of all three correct products	
		ii	$= \frac{619}{4000} = 0.15475$	A1	CAO <u>Examiner's Comments</u> Most candidates found the correct three products and calculated them correctly. A small number failed to find all three. For those who got the tree diagram wrong, follow through marks were available.	
		iii	P(Hits with exactly one given hits with at least one) $= \frac{P(\text{Hits with exactly one and hits with at least one})}{P(\text{Hits with at least one})}$			If answer to (B) > than answer to (A) then max M1M0A0 Both must be part of a fraction
		iii		M1	For numerator FT	
		iii	$= \frac{0.15475}{0.18775}$	M1	For denominator FT	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	=0.8242	A1	CAO <u>Examiner's Comments</u> Many of those who reached this part were successful. However, there was considerable confusion in finding the conditional probability, often with a correct denominator but a wrong numerator of P(at least one)×P(exactly one). Some candidates inverted the fraction.	Allow 0.824 or better or 619/751
		iv	P(Hits three times overall) = $(0.1 \times 0.2 \times 0.2) + (0.9 \times 0.95 \times 0.95 \times 0.05 \times 0.2 \times 0.2)$	M1	For $0.1 \times 0.2 \times 0.2$ or 0.004 or 1/250	FT their tree for all three M marks
		iv		M1	For $0.9 \times 0.95 \times 0.95 \times 0.05 \times 0.2 \times 0.2$	provided three terms in first product and six in second product. Last three probs must be $0.05 \times 0.2 \times 0.2$ unless they extend their tree
		iv	= 0.004 + 0.0016245	M1* Dep on both prev M1's	For sum of both	With no extras

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	= 0.0056245	A1	CAO <u>Examiner's Comments</u> Approximately one third of candidates were successful in this part. However many were confused. Many candidates successfully found the first product but then failed to find the second, or found additional products. Those who attempted the second product often made errors. The last three probabilities were often $0.1 \times 0.2 \times 0.2$ rather than $0.05 \times 0.2 \times 0.2$.	Allow 0.00562 or 0.00563 or 0.0056 NOTE RE OVER-SPECIFICATION OF ANSWERS If answers are grossly over-specified, deduct the final answer mark in every case. Probabilities should also be rounded to a sensible degree of accuracy. In general final non probability answers should not be given to more than 4 significant figures. Allow probabilities given to 5 sig fig. PLEASE HIGHLIGHT ANY OVER-SPECIFICATION Please note that there are no G or E marks in scoris, so use B instead
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	Because $P(L/R) \neq P(L)$	E1	If two or more methods given and only one correct, do not award the mark	Either $P(L \cap R) (= 0.099) \neq P(L) \times P(R)$, provided 0.099 in (ii)
		ii			Allow $0.45 \neq 0.15$ Examiner's Comments The majority of candidates who scored this mark showed that $P(L \cap R) = 0.099 \neq P(L) \times P(R) = 0.033$. Very few candidates gave the simplest explanation which is that $P(L R) \neq P(L)$. For the former, candidates had to quote the correct probabilities, but for the latter the symbolic representation was adequate, as the probabilities were given in the question.	or $0.099 \neq 0.15 \times 0.22 (= 0.033)$ Look out for complement methods, etc
		ii	$P(L \cap R) = P(L R) \times P(R) = 0.45 \times 0.22$	M1	For product	Allow if done correctly in part(i)
		ii	$= 0.099$	A1	CAO Examiner's Comments There were three common answers here. The majority correctly obtained 0.099, but some candidates multiplied the wrong probabilities together to obtain 0.033 or 0.0675. Brief working was generally given both for the correct and the incorrect answers	Allow 99/1000

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		G1	For two labelled intersecting circles, provided no incorrect labelling.	Condone labels such as $P(L)$ etc Allow other shapes in place of circles No need for 'box'
		iii		G1	For at least 2 correct probabilities. FT their $P(L \cap R)$ from part (ii) provided ≤ 0.15	FT from 0.033 in (ii) gives 0.117, 0.033, 0.187, 0.663 In general $0.15 - x$, x , $0.22 - x$, $0.63 + x$ May also see 0.0825, 0.0675, 0.1525, 0.6975
		iii		G1	For remaining probabilities. FT their $P(L \cap R)$ providing probabilities between 0 and 1. Examiner's Comments Most candidates gained full credit here, often from a follow through of a wrong answer to part (ii). Some candidates failed to subtract $P(L \cap R)$ away from $P(L)$ and $P(R)$ and but were still able to score one mark for the two labelled circles.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i				Allow labels such as A, R, F(Fail) etc
		i				All probabilities correct
		i		G1	First column	All probabilities correct
		i		G1	Second column	All probabilities correct
		i		G1	Final column	If any labels missing or incorrect allow max 2/3 Do not allow misreads here as all FT (eg 0.3 and 0.5 reversed)
Do not award if first two branches missing Branches two and three should come out of 'retest' Examiner's Comments The vast majority of candidates were able to correctly construct the tree diagram although it did appear that quite a few needed two attempts (it looked as though there made been some rubbing out under the final version). Only a very small number of candidates omitted any of the required labels or mixed up some of the probabilities, but these candidates were able to gain follow through marks in subsequent parts of the question. A few candidates omitted the middle set of branches, or added extra sets following 'Accept' or 'Reject'.						

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$P(\text{Accepted}) = 0.2 + (0.3 \times 0.2) + (0.3 \times 0.3 \times 0.4)$	M1	For second or third product	FT their tree provided correct numbers of terms and correct structure of 3, 3, 2 branches.
		iii	$= 0.2 + 0.06 + 0.036 = 0.296$	A1	CAO Examiner's Comments This was generally very well answered.	Allow 37/125 oe
		iv	$P(\text{At least one retest given accepted})$ $= \frac{P(\text{At least one retest and accepted})}{P(\text{Accepted})}$	M1	For numerator	FT their tree provided correct numbers of terms and correct structure of 3, 3, 2 branches. for both M1's
		iv	$= \frac{(0.3 \times 0.2) + (0.3 \times 0.3 \times 0.4)}{0.296} = \frac{0.096}{0.296}$	M1	For denominator	Both must be part of a fraction Allow 12/125 oe
		iv	$= 0.324$	A1	FT their 0.296 and 0.096 Allow 0.32 with working Examiner's Comments Candidates found this part much more difficult and many gave an answer of 0.096, which is simply the probability that a candidate for the job is retested at least once and accepted, so not a conditional probability at all. This scored zero unless it was as the numerator of a fraction. Other candidates did have a fraction with the correct denominator but their numerator was incorrect.	Allow 12/37 oe
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	Inter-quartile range = $18.1 - 17.8 = 0.3$	B1		
		i	Lower limit $17.8 - (1.5 \times 0.3) (= 17.35)$	M1		FT their IQR for M marks only
		i	No outliers at lower end.	A1	dep on 17.35	Allow 'No values below 17.35 for first A1 Allow 'Lower limit = 17.35 so no outliers (at lower end)' Watch for use of median giving 17.45 which gets M0A0 You must be convinced that comments about no outliers refer to lower tail only.
		i	Upper limit $18.1 + (1.5 \times 0.3) (= 18.55)$	M1		
		i	(Max is 18.6) so at least one outlier at upper end.	A1	dep on 18.55	Allow 'At least one value above 18.55' for second A1 Allow 'any value above 18.55 is an outlier' so at least one outlier. Do not allow 'There MAY be one outlier' oe Condone 'one outlier' Condone 'there are outliers' Watch for use of median giving 18.35 which gets M0A0 You must be convinced that comments about some outliers refer to upper tail only.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i			Examiner's Comments Most candidates calculated the inter-quartile range and used it correctly to find the limits for outliers. However a few used the median instead of the quartiles to add to and subtract from $1.5 \times$ inter-quartile range. Many candidates neglected to comment on outliers at each end separately (only commenting if there were outliers overall).	
		ii	$P(A) = P(\text{All 3 have orange centres}) =$ $\frac{7}{20} \times \frac{6}{19} \times \frac{5}{18} = \frac{7}{228}$	M1	For $7/20 \times$	Allow final answer of 0.031 with working
		ii		M1	For product of correct three fractions Without extra terms	ALTERNATIVE SCHEME ⁷ $C_3/^{20}C_3$ $= 35/1140 = 7/228 = 0.0307$ M1 for either term in correct position in a fraction
		ii	$= 0.0307 \text{ (0.030702)}$	A1	CAO Allow full marks for fully simplified fractional answers	M1 for correct fraction A1 CAO
		ii	$P(B) = P(\text{All 3 have same centres}) =$ $\left(\frac{7}{20} \times \frac{6}{19} \times \frac{5}{18}\right) + \left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18}\right) +$ $\left(\frac{4}{20} \times \frac{3}{19} \times \frac{2}{18}\right) + \left(\frac{3}{20} \times \frac{2}{19} \times \frac{1}{18}\right) =$	M1	For at least two correct triple products or fractions or decimals	ALTERNATIVE SCHEME ⁷ $C_3/^{20}C_3 + {}^6C_3/^{20}C_3 + {}^4C_3/^{20}C_3 + {}^3C_3/^{20}C_3$ M1 for at least two correct terms
		ii	$= 0.0307 + 0.0175 + 0.0035 + 0.0009$	M1	For sum of all four correct	M1 for sum of all four (all correct) either as combinations or decimals
		ii	$= 0.0526 = \frac{1}{19} \text{ (0.052632)}$	A1	CAO	A1 CAO

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\left(= \frac{7}{228} + \frac{1}{57} + \frac{1}{285} + \frac{1}{1140} \right)$		Allow 0.053 or anything which rounds up to 0.053 with working Examiner's Comments This question was answered well (even by candidates who had struggled with earlier questions). Many answers were left as fractions (which were exact) with very few marks lost for over-specification. However many candidates squashed their work up into the first part of the answer space, not realising that there was more space on the next page. A few candidates forgot to take into consideration 'non replacement' and starting with $(7/20)^3$, gained only 1 mark, although plenty of follow through marks were available in parts (iii) and (iv).	Please check all of the answer space for this part
		iii	$P(A B) = \frac{0.0307..}{0.0526..}$	M1	For their 'A' divided by their 'B'	Allow 0.584 from $\frac{0.0307}{0.0526}$
		iii	$= 0.583 (= 0.58333)$	A1	FT their answers to (ii) provided answer < 1	$\frac{7}{12}$ Allow $\frac{7}{12}$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$P(B A) = 1$	B1	CAO Examiner's Comments Although over half of candidates gained at least 2 marks out of 3 here, a number of candidates did not use the straightforward $P(A B) = P(A)/P(B)$ here, and instead mistakenly calculated $P(A \cap B) = P(A) \times P(B)$, and then cancelled out a term on the top and bottom of the fraction. This illustrates the lack of deep understanding here of independence and conditionality. A disappointing number of candidates were quite happy to give an answer greater than 1 for $P(B A)$.	
		iv	$P(\text{All have orange centres})$ $= 0.0307^2 = 0.00094$ $\text{or } = \frac{49}{51984}$	M1	For their 0.0307^2	Allow 9.4×10^{-4} condone 0.0009 or 9×10^{-4}
		iv	$= (0.00094260)$	A1	FT Examiner's Comments This was well-answered although a common incorrect method was $2 \times P(A)$.	
		v	$P(\text{Has to select } > 2) = 1 - P(\text{Has to select } \leq 2)$ $= 1 - \left(\frac{14}{20} + \left(\frac{6}{20} \times \frac{14}{19} \right) \right)$ $= 1 - (0.7 + 0.221) = 1 - 0.921$	M1	For $\left(\frac{6}{20} \times \frac{14}{19} \right)$	For any of the methods below allow SC2 for $1 - 0.079 = 0.921$ or $1 - 3/38$ $= 35/38$ o.e. as final answer
		v		M1	For $1 - \text{sum of both}$	This is $1 - P(C' + CC')$
		v	$= 0.079 (=0.078947)$	A1	CAO	
		v	OR			

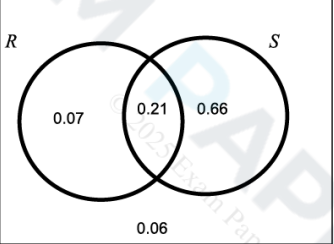
Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	$P(\text{Has to select } > 2) =$ $P(\text{First 2 both cherry}) =$ $\left(\frac{6}{20} \times \frac{5}{19}\right)$	M2	For whole product	Without extra terms added M1 if multiplied by $k/18$ only where $0 < k < 18$ (seen as a triple product only)
		v	$= 0.079 = \frac{3}{38}$	A1	CAO	This is $P(CC)$.
		v	OR			
		v	$1 - (P(0 \text{ cherries}) + P(1 \text{ cherry})) =$ $1 - \left(\frac{14}{20} \times \frac{13}{19} + \left(\frac{6}{20} \times \frac{14}{19}\right) + \left(\frac{14}{20} \times \frac{6}{19}\right)\right)$ $= 1 - (0.4789 + 0.2211 + 0.2211) = 1 - 0.9209$	M1	For any term	
		v		M1	For $1 - \text{sum of all three}$	This is $1 - P(C'C' + CC' + C'C)$
		v	$= 0.079$	A1	CAO	
		v	OR			
		v	$\left(\frac{6}{20} \times \frac{5}{19} \times \frac{14}{18}\right) + \left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{14}{17}\right) +$ $\left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{3}{17} \times \frac{14}{16}\right) + \left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{3}{17} \times \frac{2}{16} \times \frac{14}{15}\right) +$ $\left(\frac{6}{20} \times \frac{5}{19} \times \frac{4}{18} \times \frac{3}{17} \times \frac{2}{16} \times \frac{1}{15} \times \frac{14}{14}\right)$	M1	For any term	This is $P(CCC' + CCCC' + CCCCC' + CCCCCC' + CCCCCC')$
		v	$= \frac{7}{114} + \frac{14}{969} + \frac{7}{2584} + \frac{7}{19380} + \frac{1}{38760}$	M1	For sum of all five terms (all correct)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	=0.079	A1	CAO Examiner's Comments Only approximately a quarter of candidates produced a completely correct answer, although many went through the correct answer on the way to a wrong one. Most of the correct answers used the $6/20 \times 5/19$ method, with a large minority then compromising this with another term multiplied or added.	
			Total	19		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	$P(\text{Neither is an ace}) = \left(1 - \frac{4}{52}\right)^2$	M1	For 48/52 oe seen	
		i	$= \frac{2304}{2704} = \frac{144}{169} = 0.852 \text{ (0.8572071...)}$	A1	CAO Examiner's Comments This was answered very well. However a few ignored the question and assumed 54 or 50 cards in a pack, or that the card had not been replaced. Another common wrong method was to find $1 - P(\text{both aces}) = 1 - (4/52)^2 = 168/169$.	Allow 0.85 with working
		ii	Expected number = $10 \times 0.852 = 8.52$	B1	FT their (i) if seen Examiner's Comments Surprisingly, only two thirds of candidates scored this easy mark. Most realised that they had to multiply their answer to part (i) by 10 but some then rounded their answer to a whole number, thus losing the mark. A smaller number incorrectly raised their answer to part (i) to the power of 10.	Do not allow whole number final answer even if 8.52 seen first. Allow fractional answer
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
19		i	(A) P(Watched cyc but not fb) = $\frac{15}{250} = \frac{3}{50} = 0.06$	B1	CAO (aef) Examiner's Comments This was answered very well, although a number of candidates gave the number that watched cycling and not football rather than the probability. A few had the wrong divisor, usually 186 or 100.	
		i	(B) P(Watched one or two) = $\frac{33+12+21+14+3+65}{250}$	M1	OR: $\frac{250 - (64 + 38)}{250} =$	For M1 terms must be added with no extra terms (added or subtracted)
		i	$= \frac{148}{250} = \frac{74}{125} = 0.592$	A1	CAO (aef) Examiner's Comments Again this was very well-answered with only a small minority of candidates making errors. The most common errors were to include those people who watched all three sports or to miss out one of the six who watched 1 or 2 sports..	
		ii	P(Not watched fb watched cyc) = $\frac{15}{67} = 0.224$ (0.223880597...)	M1		For denominator of either 67 or 67/250 or 0.268

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	CAO (aef) Examiner's Comments Approximately two thirds of candidates answered this correctly. Of the rest, some were able to get a method mark for the correct denominator but then failed to get the correct numerator, often thinking that it was 12 rather than 15. Some candidates either did not recognize this as a conditional probability question, or did not know about conditional probability.	Allow 0.22 with working
			Total	5		

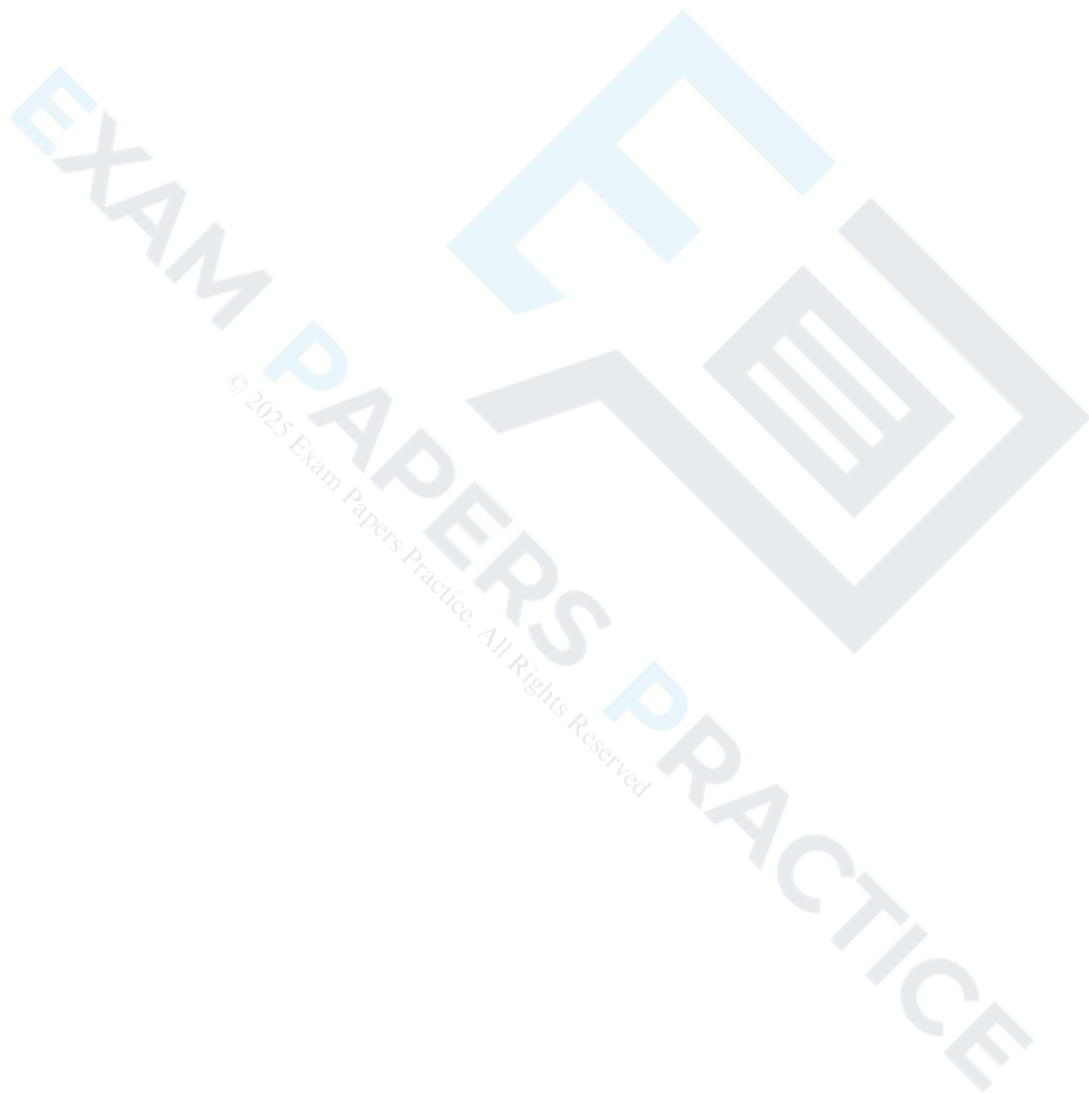
Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	$P(R \cap S) = P(R) + P(S) - P(R \cup S)$ $= 0.28 + 0.87 - 0.94$ $= 0.21$	M1	For correct use of formula	Or $0.28 - x + 0.87 - x + x = 0.94$
		i		A1	Examiner's Comments This part was very well answered but a considerable number of candidates assumed that the probabilities were independent and calculated $P(R) \times P(S)$. Some were more confused about the correct formula to use and calculated $P(R) \times P(R \cup S)$.	
		i				
		ii		G1	For two labelled intersecting circles	Allow labels such as $P(R)$ and $P(S)$ Allow other sensible shapes in
		ii		G1	For at least 2 correct probabilities. FT their $P(R \cap S)$	place of circles Allow their $P(R \cap S)$ rounded to 2dp For both G1 marks FT their 0.21

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		G1	For remaining probabilities. FT their $P(R \cap S)$ Examiner's Comments The idea of the Venn diagram was well understood, and most candidates produced a fully correct solution (often following through from an error in part (i)). Very few noticed the contradiction produced by their wrong answer, which gave the outer zone as 0.0936 instead of 0.06 from the question.	provided < 0.28 For FT if $P(R \cup S) = x$ then others are $0.28 - x$, $0.87 - x$, $x - 0.15$ 0.2436 leads to 0.0364, 0.6264, 0.0936
		iii	$P(R S) = \frac{P(R \cap S)}{P(S)} = \frac{0.21}{0.87} = \frac{21}{87} = 0.241$	M1	for fraction	Allow $\frac{7}{29}$ or $\frac{21}{87}$ as final answer
		iii	Exact answer 0.241379...	A1	CAO FT their part (i) (for M1 only) but M0 if their answer to part (i) is $P(R) \times P(S)$	Allow 0.24 with working Condone 'if' or 'when' for 'given that' but not the words 'and' or 'because' or 'due to' for E1.

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	This is the probability that (on a randomly selected day) there is at least 1 mm of rain, given that there is at least 1 hour of sun.	E1	Need more than just probability of rain given sun Must include 'probability' or 'chance' oe Do not allow just P(at least 1 mm of rain, given that there is at least 1 hour of sun) Examiner's Comments Among those who had not made the independence error in part (i), the correct answer was quite common. The explanation of what the probability means was usually correct but sometimes lacked sufficient detail. There were a few candidates who 'reversed' the statement and gave an explanation of $P(S R)$.	E1 (independent of M1): the order / structure must be correct i.e. no reverse statement Allow 'The probability that on a randomly selected day when there is at least 1 hour of sun there is at least 1 mm of rain.' oe
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	Number of ways = $5! = 120$	B1	Examiner's Comments This was generally well answered.	
		ii	Probability = $2/120$	M1	For division by their 120	M1 for $^k/_{120}$
		ii	$= 1/60$ or 0.0167 or 0.01 6	A1	CAO Examiner's Comments Again this was generally well answered, although some candidates truncated their decimal rather than correctly rounding. The use of fractions was preferable here.	Condone final answer of $2/120$ Do not allow 0.016
		iii	$\frac{1}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{1}{60}$ or 0.0167 or 0.01 6	B1	Condone $2/120$ for B1 Examiner's Comments Only around two thirds of candidates scored the mark here. The most common error was to use combinations.	Do not allow 0.016
		iv	$\frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{10}$	M1	For $3/5 \times$	Listing options gives $12/120$
		iv		A1	CAO	Or $(3! / 120) \times 2$ Or $(^3P_3 \times ^2P_2)/120$
		iv	or $1/5 C_3 = 1/10$	M1	For division by 5C_3 Examiner's Comments Again about two thirds of candidates scored both marks, with many scoring the marks for their answer to part (iii), correct or not, multiplied by 6, rather than for the fairly simple $^3/5 \times ^2/4 \times ^1/3$.	SC2 for $3! \times$ their part (iii) or $6 \times$ their part (iii)
		iv		A1		

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	6	


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Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	$P(\text{Does not lose any match}) = 0.83 = 0.512 = \frac{64}{125}$	B1	Examiner's Comments Many candidates thought that not losing meant winning, and hence gave the common wrong answer of $0.5^3 = 0.125$. Others tried to consider combinations of wins and draws, often without success. The fact that the question part was only worth 1 mark should have been a clue to the fact that there was an easier approach.	
		ii	$P(\text{Wins all 3 or draws all 3 or loses all 3}) = 0.5^3 + 0.3^3 + 0.2^3$	M1	Including addition Examiner's Comments This was generally well answered although a few candidates interpreted this as 'find three separate probabilities which they did, and listed them but with no addition thus scoring zero.	
		ii	$= 0.16 = \frac{4}{25}$	A1		
		iii	$P(\text{all three outcomes occur}) = 3! \times 0.5 \times 0.3 \times 0.2$	M1*		Allow M1 for $k \times 0.5 \times 0.3 \times 0.2$ even if $k = 1$ Even if cubed
		iii	$= 0.18$	A1		
		iii	Required probability = $1 - '0.18' - 0.16$	*M1 dep		Not if cubed
		iii	$= 0.66 = \frac{33}{50}$	A1	CAO	
		iii	OR:			
		iii	$P(\text{WWW}) + P(\text{DDD}) + P(\text{LLL})$	M1	For any one product (no need for '3 ×')	Even if cubed
		iii	$3 \times 0.5^2 \times 0.5 + 3 \times 0.3^2 \times 0.7 + 3 \times 0.2^2 \times 0.8$	M1	For '3 ×'	Dep on at least 1 correct term

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	0.375 + 0.189 + 0.096	M1	For sum of three correct terms (no need for '3 ×') And no incorrect terms	NB common wrong answer of 0.22 from omitting '3 ×' or 0.44 from '2×' scores M1M0M1A0 Not if cubed
		iii	0.66	A1	CAO	
		iii	OR:			
		iii	P(WWD) + P(WWL) + P(DDW) + P(DDL) + P(LLW) + P(LLD)	M1	For any one product (no need for '3 ×')	Even if cubed
		iii	$3 \times 0.5^2 \times 0.3 + 3 \times 0.5^2 \times 0.2 + 3 \times 0.3^2 \times 0.5 + 3 \times 0.3^2 \times 0.2 + 3 \times 0.2^2 \times 0.5 + 3 \times 0.2^2 \times 0.3$	M1	For '3 ×'	Dep on at least 1 correct term
		iii	0.225 + 0.15 + 0.135 + 0.054 + 0.06 + 0.036	M1	For sum of six correct terms (no need for '3×') And no incorrect terms	Not if cubed
		iii	0.66	A1	CAO	
					Examiner's Comments Only a small proportion of candidates used the most elegant approach (the first method in the mark scheme) and of those who did, many forgot to multiply by 6. Most candidates gave lists or tree diagrams to show P(WWL) etc., but many then did not multiply by 3, so the most common answer was 0.22, rather than the correct 0.66.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23			$P(A \cap B) = P(A) + P(B) - P(A \cup B)$ $= 0.6 + 0.5 - 0.85$ $= 0.25$ $P(A B) = \frac{P(A \cap B)}{P(B)}$ $= \frac{0.25}{0.5}$ $= 0.5$	M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1(AO1.1) [4]		
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
24		a	$1 - 0.51^2$ $= 0.7399$	M1(AO3.1b) A1(AO1.1) [2]	Accept 0.74 or 0.740		
		b	$1 - 0.49^2 - 0.38^2 - 0.1^2 - 0.03^2$ $= 0.6046$	M1(AO3.1b) M1(AO1.1) A1(AO1.1) [3]	For squaring probabilities OR products of pairs For complementary event OR doubling products of pairs $2 \times (0.1862 + 0.049 + 0.0147 + 0.038 + 0.0114 + 0.003)$		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
25		a	0.4×0.6 or 0.6×0.1 $p(\text{in, out}) + p(\text{out, out})$ 0.3	B1(AO 3.1b) M1(AO 3.1a) A1(AO 1.1) [3]	If B0M0 allow SC1 for correct tree diagram with all possible outcomes shown.		
		b	$1 - p(\text{out every night})$ $1 - 0.6 \times 0.1^6$ $= 0.9999994 \approx 1$ so extremely likely	M1(AO 3.1b) A1(AO 1.1) A1(AO 3.2a) [3]			
			Total	6			
26			$p(\text{G or C}) [= 1 - 0.0985] = 0.9015$ $P(\text{G and C}) = 0.901 + 0.008 - 0.9015 [= 0.0075]$ $P(\text{C/G}) = \frac{\text{their } 0.0075}{0.901}$ $= 0.00832$ This is very unlikely so he is incorrect.	B1(AO 3.1a) M1(AO 1.1) M1(AO 2.1) A1(AO 1.1) E1(AO 3.2a) [5]	$P(\text{G or C}) = P(\text{G}) + P(\text{C}) - P(\text{G and C})$ used		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i	$\binom{50}{4} = \frac{50!}{4!46!} = 230300$	M1 A1 [2]	Examiner's Comments This part was answered well by the majority of the candidates. A small minority of the candidates used permutations instead of combinations and a very few simply gave an answer of $4! = 24$.	
		ii	$\frac{17}{50} \times \frac{16}{49} \times \frac{15}{48} \times \frac{14}{47} = \frac{17}{1645} = 0.0103$ Or: $\binom{17}{4} \div \binom{50}{4} = \frac{2380}{230300} = 0.0103$	M1 A1 [2] M1 A1	$17/50 \times$ or $0.34 \times$ NB $\left(\frac{17}{50}\right)^4$ or 0.34^4 scores M1A0 But M0 if part of a binomial expression CAO Uncancelled fraction gets M1A0 Uncancelled fraction gets M1A0 Allow 0.010 with working but not 0.01 Examiner's Comments This was answered well by many candidates. However a common misconception was to assume 'with replacement' probability giving an answer of $0.34^4 = 0.0134$. A few candidates gave their answer in fractional form but failed to cancel, hence losing the accuracy mark.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$1 - 4! \times \frac{17}{50} \times \frac{9}{49} \times \frac{13}{48} \times \frac{11}{47}$ $= 1 - 24 \times 0.0003958..$ $= 1 - 0.09500217 = 0.905$ $(0.904997...)$ Or: $1 - \left[\frac{\binom{17}{1} \times \binom{9}{1} \times \binom{13}{1} \times \binom{11}{1}}{\binom{50}{4}} \right] = 1 - 0.09500 = 0.905$	M1 M1 M1 A1 [4]	For correct product For x4! For 1- with product of four fractions but with or without a coefficient CAO If denominators all 50 then max M0M1M1A0 Allow 0.90 with working	
				M1 M1 M1 A1	For product of four correct nCr terms For division of product of four nCr terms by 50C4 For 1 - product of four nCr terms divided by 50C4	
					Examiner's Comments This question was found to be rather difficult. Many candidates wrote more than one page as they attempted to find all the probabilities of two or more using the same method, almost always without success. In fact around half of the candidature scored zero on this question part. A fair number of candidates did find the correct product and took their answer away from 1, but few found the correct multiplier of 4!, and thus only gained 2 marks.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	Probability = $(1 - 0.4) \times 0.8 = 0.48 \left(= \frac{12}{25} \right)$	M1 M1 [2]	Examiner's Comments Most candidates found the correct probability although a few did incorrectly find $0.4 \times 0.8 = 0.32$ or $0.4 \times 0.2 = 0.08$.	
		ii	Either: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= 0.4 + 0.8 - 0.4 \times 0.8$ $= 0.88 \left(= \frac{22}{25} \right)$ Or: $P(A \cup B) = 0.4 \times 0.8 + 0.6 \times 0.8 + 0.4 \times 0.2$ $= 0.32 + 0.48 + 0.08$ $= 0.88$ Or: $P(A \cup B) = 1 - P(A' \cap B')$ $= 1 - 0.6 \times 0.2 = 1 - 0.12 = 0.88$	M1 M1 A1 CAO [3] M1 M1 A1 M1 M1 A1	for use of formula for 0.4×0.8 For any two terms For all three terms For 0.6×0.2 For complete method Examiner's Comments This part was again very well answered. Many candidates gave the correct method and answer. A minority of the candidates gave an answer of 0.56 by adding two terms rather than three terms. These candidates missed out the 0.32 term.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$P(A B) = \frac{P(A \cap B)}{P(B)} = \frac{0.32}{0.88} = \frac{4}{11} = 0.364 = (0.3636\dots)$	M1 M1 A1FT [3]	For numerator As part of fraction (with denominator) For denominator As part of fraction (with numerator) FT their answer to part (ii) Accept 0.36 (dots above) Accept 0.36 with correct working Examiner's Comments It is pleasing to report that this conditional probability question was well-answered by many candidates, two thirds whom gained full credit. Most recognised that they had to divide by the probability found in part (ii) although a few did not use the correct numerator. Those who found the correct probability sometimes lost the accuracy answer for poor rounding, giving, for example, a final answer of 0.366 rather than 0.364.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i	$0.55^3 = 0.166 \quad (0.166375) \left(= \frac{1331}{8000} \right)$	M1 A1 [2]	For 0.55^3 Accept 0.17 with working Condone answer of 0.166375 (over-specified)	
					Examiner's Comments Around 95% of candidates answered this correctly. Most gave a decimal answer of 0.166 or 0.1664.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$P(\text{Wins in 3 games}) = 0.45^3$ $= 0.091125$ $P(\text{Wins in 4 games}) = 3 \times 0.45^2 \times 0.55 \times 0.45 = 0.150356$ $P(\text{Wins in 5 games}) = 6 \times 0.45^2 \times 0.55^2 \times 0.45 = 0.165392$ NB Answer if no coefficients used is 0.168809... $= 0.091125 + 3 \times 0.05011875 + 6 \times 0.0275653125$ $P(\text{Emily wins}) = 0.407$ (0.406873..) $1 - P(\text{Sakura wins})$ can get all marks (use similar scheme) so eg $1 - 0.55^3$ gets M1M0M0M0A0	M1 M1 M1 M1 A1 [5]	For $P(\text{Wins in 3 games})$ $P(\text{Wins in 4 games})$ with any or no coefficient $P(\text{Wins in 5 games})$ with any or no coefficient For either coefficient correct CAO SC2 for $P(\text{Sakura wins}) = 0.593$ (0.593126...) Examiner's Comments This caused numerous problems for many candidates – time was spent drawing tree diagrams and trying to list all combinations. Most candidates found the probability of Emily winning in 3 games, 0.45^3, and thus gained a mark. A large number also found the probability of her winning in 4 games (in some order) and in 5 games (in some order). The number of valid combinations was frequently wrong – sometimes failing to find them all systematically and often using 4C_3 and 5C_3. Some candidates used $1 - P(\text{Sakura wins})$ and these candidates achieved mostly between 1 and 3 marks with mistakes with the coefficients resulting in not gaining full credit. An elegant solution seen was from a candidate who stated that Emily must win the last game and so worked on the possible ways of Emily winning two of the previous matches.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	7		
30			$P(\text{correct}) = \frac{2}{5} + \frac{3}{5} \times \frac{1}{3}$ $= 0.6 \left(\frac{3}{5}\right)$ $\text{their } 0.6^2$ $0.36 \left(\frac{9}{25}\right)$	M1(AO3.1b) A1(AO1.1b) M1(AO1.1b) A1(AO1.1b) [4]		
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
31			$[0.8 \times 0.2 =] 0.16$ seen 0.35×0.5 and 0.45×0.1 seen Finds sum of their attempt at these probabilities 0.38 <i>their</i> $0.16 \div$ <i>their</i> 0.38 $\frac{8}{19}$	B1(AO1.1b) B1(AO3.1b) M1(AO3.1a) A1(AO2.1) M1(AO1.1b) A1(AO3.1a) [6]	or tree diagram with all necessary outcomes and associated probabilities may be implied by 0.38 0.42105263... rounded to 3 sf or more	allow omission of 0.2, 0.5 and 0.9 together with labelling	
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
32		a	$0.2 + 0.15 + 0.3 + k + 0.25 = 1$ $k = 0.1$	M1(AO1.1) A1(AO1.1) [2]	Examiner's Comments Candidates generally did this correctly, with many showing exactly the working given in the mark scheme.	
		b	k^2 seen $0.3 \times 0.25 \times 2 + k^2$ 0.16	M1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	Or $0.3 \times 0.25 \times 2$ seen Allow slip eg omission of 2; or insertion of 2 Ft k from (a) for M1M1 Examiner's Comments This part caused considerable difficulty, with some candidates omitting it, others drawing a possibility space with equally likely outcomes, and others copying information from the table given in the question. Candidates who realised that a total of 6 could be made either from 3 and 3 or else from 2 and 4 sometimes did not realise that 4 and 2 is also possible; others thought the probability of 3 and 3 had to be counted twice.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
33		a	0.36	B1 (AO 1.1) [1]	<u>Examiner's Comments</u> A small minority of candidates found the probability to win the next two league football matches.	
		b	$P(\text{draw}) \neq 0$ oe	B1 (AO 2.4) [1]	allow any comment which identifies that other outcomes are possible eg winning and losing are not exhaustive <u>Examiner's Comments</u> Generally answered well, although some mini essays seen.	
			Total	2		

Question		Answer/Indicative content	Marks	Part marks and guidance	
34		use of contingency table or Venn diagram or $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ $P(A \text{ and } B) = 0.5$ $P(A) \times P(B) = 0.56 \times 0.80$ $= 0.448$ seen $0.448 \neq 0.5$ or $0.56 \times 0.80 \neq 0.5$ so not independent	M1 (AO 3.1b) A1 (AO 2.1) M1 (AO 1.1) A1 (AO 1.1) A1 (AO 3.2a) [5]	0.56, 0.8 and 0.14 must be correctly placed; eg $1 - 0.14 = 0.56 + 0.8 - P(A \text{ and } B)$ or $P(A/B) = \frac{0.5}{0.80}$ $= 0.625$ or $\frac{5}{8}$ seen 0.625 or $\frac{5}{8} \neq 0.56$	where A denotes “passing” maths and B denotes “passing” English the first M1A1 may be awarded for working with percentages allow equivalent argument based on showing A' and B' not independent
				Examiner's Comments Candidates who did well in this question defined appropriate events and set out a clear, reasoned argument. They usually worked with a Venn diagram and clearly showed that $p(A \text{ and } B) \neq p(A) \times p(B)$ in this case. Candidates who did less well wrote down relevant ideas but were unable to draw them together successfully. They sometimes stated results instead of giving details of the calculation. Exemplar 1	

Question			Answer/Indicative content	Marks	Part marks and guidance
					Let $P(A)$ be probability of getting ≥4 in maths Let $P(B)$ = probability of getting ≥4 in English $P(A) = 0.56$ If independent $P(A \cap B) = P(A) \times P(B)$ $P(B) = 0.8$ $P(A \cap B) = 0.448$ $P(A) \times P(B) = 0.448$ ✓✓ This candidate has been given BOD M1A1 for the calculation of $p(A) \times p(B)$ being correctly done, as highlighted by the statements on the LHS. The events are clearly defined. No attempt was made to find the actual value of $p(A$ and $B)$, so no further progress was made.
			Total	5	

Question		Answer/Indicative content	Marks	Part marks and guidance		
35		$\frac{15}{n} \times \frac{14}{n-1} + \frac{n-15}{n} \times \frac{n-1-15}{n-1} = 2 \times \frac{15}{n} \times \frac{n-15}{n-1}$ Multiply through to obtain quadratic in n $n^2 - 61n + 900 = 0$ $n = 36$ (not 25) since more red discs	M1 (AO3.1b) M1 (AO2.1) A1 (AO1.1) M1 (AO2.1) A1 (AO1.1) B1 (AO3.2b)	P(BB) + P(RR) = P(one of each) Two of three terms correct All correct	n is the total number of discs	
		$\frac{15}{36} \times \frac{14}{35} + \frac{21}{36} \times \frac{20}{35}$ $\frac{15}{36} \times \frac{14}{35} + \frac{21}{36} \times \frac{20}{35}$ $\frac{1}{3}$ OR $\frac{15}{15+r} \times \frac{14}{14+r} + \frac{r}{15+r} \times \frac{r-1}{14+r} = 2 \times \frac{15}{15+r} \times \frac{r}{14+r}$ Multiply through to obtain quadratic in r $r^2 - 31r + 210 = 0$ $r = 21$ (not 10) since $r > 15$ $\frac{15}{36} \times \frac{14}{35} + \frac{21}{36} \times \frac{20}{35}$ $\frac{1}{3}$	M1 (AO2.1) A1 (AO1.1) [8] M1 M1 A1 M1 A1 B1 M1 A1 [8]	so there are 21 red discs FT their 44 Accept decimal equivalent to 2 or more dp P(BB) + P(RR) = P(one of each) Two of three terms correct All correct FT their 21 (> 15) Accept	r is the number of red discs	

Question			Answer/Indicative content	Marks	Part marks and guidance		
					decimal equivalent to 2 or more dp		
			Total	8			
36		a	$\frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.6} = \frac{1}{3}$	B1(AO 1.1) [1]	Accept use of Venn diagram		
		b	$P(A \cup B) = 0.9$ $P(A \cap B') = 0.9 - 0.6 = 0.3$ $\frac{0.3}{0.4} = 0.75$	B1(AO 2.1) B1(AO 1.1) B1(AO 1.1) [3]	Accept use of Venn diagram		
		c	$\frac{1}{3} \neq \frac{3}{4}$ or $\frac{1}{3} \neq \frac{1}{2}$ or $\frac{3}{4} \neq \frac{1}{2}$ so not independent	B1(AO 2.2a) [1]			
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
37		a	eg median $\approx$ mean eg extreme values and/or quartiles symmetrical about mean oe	B1 (AO 2.4) [1]		
		b	$P(X < 46.4) = 0.24956... \approx 0.25$ $P(X < 66.1) = 0.75262... \approx 0.75$ So both values are consistent	M1 (AO 3.3) A1 (AO 2.4) [2]	BC Must follow from correct calc. values	or invNorm(0.25, 56.2, 14.5) = 46.4199... $\approx$ 46.4 i nv Norm(0.75, 56.2, 14.5) = 65.98... $\approx$ 66.1
		c	0.00987574855 ...	B1 (AO 1.1) [2]	BC to 3 or more sf	
		d	Use of $N(112.4, 29^2)$ soi 0.780065376569...	M1 (AO 3.1a) A1(AO 1.1) [2]	BC to 3 or more sf	
		e	thei $\frac{0.78006...}{0.00987... + 0.78006...}$ r 0.98750...so approximately a 99% chance the phone is Wilsons so their father's statement is justifiable	M1 (AO 2.1) A1(AO 2.2b) [2]	their $\frac{0.00987...}{0.00987... + 0.78006...}$ 0.0125...so approximately a 1% chance the phone is Chidera's so their father's statement is justifiable	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance		
38		a	$\frac{1}{24} + \frac{23}{24} \times \frac{1}{23} + \frac{23}{24} \times \frac{22}{23} \times \frac{1}{22}$ $= \frac{1}{8}$	M1 (AO 2.1) M1(AO 2.4) M1(AO 1.1) [3]	$\frac{1}{24}$ seen addition of three terms		
		b	Use $B(24, \frac{1}{8})$ of 0.139(009338...) to 3, 4 or 5 sf	M1 (AO 3.3) M1(AO 1.1) [2]			
		c	$\left(\frac{1}{8}\right)\left(\frac{7}{8}\right)^{23}$ $(0.0057955\dots)^2 = 0.000033588489\dots$ to 3, 4 or 5 sf	M1 (AO 3.1a) A1(AO 1.1) [2]	soi NB 0.00 57955....		
		d	$[6000 \div 120] = 50$ $[1 - 0.13900\dots]^{49} \times 0.0057955\dots$ 0.00000376653... so Mark is wrong – it's more like 4 in a million	B1 (AO 3.1b) M1(AO 3.1a) A1(AO 1.1) A1(AO 3.2a) [4]	FT their part (b)		
			Total	11			

Question		Answer/Indicative content	Marks	Part marks and guidance		
39	a	0.4×0.4 or 0.6×0.2 seen $0.4 \times 0.4 + 0.6 \times 0.2$ 0.28	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) [3]	or tree diagram with correct outcomes and probabilities shown	allow $0.4 \times 0.4 \times 0.4$ or $0.4 \times 0.6 \times 0.2$ 2 mark the final answer	
<u>Examiner's Comments</u> Candidates who did well in this question realised the need to work with P(LW) and P(WW). Many successfully went on to earn all 3 marks, although a few made slips with the arithmetic. Candidates who did not do well incorporated the probability of winning in week 4 – often this was arbitrarily assigned the value of 0.5 and some thought they needed to work with the probabilities for the first three weeks as well.						

Question		Answer/Indicative content	Marks	Part marks and guidance		
	b					
		$\frac{1}{120}$ or $\frac{5}{120}$ seen $\frac{1}{24}$ or 0.041666...to 2 or more sf	M1 (AO3.1a) A1 (AO1.1) [2]	B2 for 0.041666... unsupported B0 for 0.042 unsupported	mark the final answer	
Examiner's Comments Candidates who did well in this question understood that the desired probability is simply $5 \times \frac{1}{120}$, although a few obtained the correct answer from working out $1 - P(\text{Alf doesn't win})$. Candidates who did less well often found $\frac{1}{24}$ successfully, but then went on to use it in the binomial distribution.						

Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$1 - \text{their } \frac{1}{24} \text{ evaluated}$ $1 - \left(\frac{23}{24}\right)^n > 0.95 \text{ FT}$ $\left(\frac{23}{24}\right)^n < 0.05 \text{ FT}$ $n = 71 \text{ cao}$	M1 (AO3.1a) M1 (AO2.1) M1 (AO1.1) A1 (AO2.2a) [4]	allow use of =, ≥ or ≤ allow use of = or ≤ 70.3890... unsupported but rounded to 1 or more dp implies M1M1M1	award full marks for 71 unsupported or from trial and improvement	
					Examiner's Comments Candidates who did well in this question subtracted their answer to the above part from 1 and then compared this value to the power of n with 0.05. Often the marks given were FT marks for working with an incorrect probability. Many used trial and improvement instead of logarithms to achieve their final answer. However, the majority of candidates were unable to make meaningful progress with this question, often working with $n \times$ their p .		
			Total	9			

Question		Answer/Indicative content	Marks	Part marks and guidance		
40	a	C: cycles B: catches bus R: raining NR :not raining	M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) [3]	correct structure outcomes correct probabilities correct Key not required accept NC: does not cycle	Accept variations on labelling if meaning clear	
Examiner's Comments The vast majority of candidates were able to draw a correct tree diagram with correct probabilities and outcomes, but a small number of candidates started with cycling or bus and then went on to raining and not raining . A small number of tree diagrams had three branches and a few others had incorrect probabilities.						

Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		0.2×0.4 or 0.8×0.9 $0.2 \times 0.4 + 0.8 \times 0.9$ 0.8 cao	M1 (AO3.1b) A1 (AO1.1) A1 (AO1.1) [3]	Ft (a) Ft (a) Accept $\frac{4}{5}$ or 80%	Accept fractions Working may be on diagram in (a)	
					Examiner's Comments This part was well done. A small number of candidates with an incorrect tree diagram in part (a) did this part correctly from the information in the question.		
	c		(their 0.8) ⁵ 0.328 or 0.3277 or 0.32768	M1 (AO2.1) A1 (AO1.1) [2]	$\frac{1024}{3125}$, accept 0.33 ; accept as percentage		
					Examiner's Comments Almost all candidates correctly evaluated $(0.8)^5$, while a few used a Binomial distribution, usually correctly. Common wrong answers came from 5×0.8 , $(0.8)^7$ or 7×0.8 .		
			Total	8			

Question		Answer/Indicative content	Marks	Part marks and guidance		
41	a	$\left[\frac{19.8}{36} =\right] 0.55$	B1 (AO1.1) [1]	or $\frac{11}{20}$ Examiner's Comments Most candidates did this correctly, with some candidates using the values 19 800 000 and 16 200 000 rather than the easier values of 19.8 and 16.2.	Must be dec or simplified frac; but condone 55%	
	b	0.0380(96943116...) BC	B1 (AO1.1) [1]	Correct to at least 2 sf Examiner's Comments Most candidates used their calculators correctly to find the answer. A common wrong answer was 0.28 (from $7 \div 25$).	Allow truncated	
	c	$H_0: p = 0.45$ $H_1: p < 0.45$ p is the probability that a driver speeding / caught speeding / on speed awareness course is female Use or sight of B(25, 0.45) $P(X = 7) = 0.06385(0773831)$ $0.06385 > 0.05$ Result is NOT significant or "accept H_0 " There is insufficient evidence (at the 5% level)	B1 (AO1.1) B1 \$(AO2.5) M1 *(AO3.3) A1 (AO1.1) *M1 (AO3.4) B1 (AO1.1) E1 (AO2.2b)	$O H_0: p = R 0.55$ $H_1: p > 0.55$ p is the probability that a driver speeding / caught speeding / on speed awareness course is male OR Use or sight of B(25, 0.55) $P(X = 18) = 1 - P(X =$	Both hypotheses Must not be inconsistent with H_1 Can be implied by $P(X = 7)$ (OR $P(X = 18) = 0.038..$	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			to suggest that drivers speeding / caught speeding / on speed awareness course are more likely to be male. oe	[7]	$17) = 1 - 0.936149 = 0.06385$ FT their probability Dep on 0.06385 Dep all previous marks except B1\$ SC: Use of $p = 0.5$ can score B1B1M1 (can be implied by 0.0216 or 0.0143) A0M1B0E0 May be incorporated in conclusion Not assertive		
					<u>Examiner's Comments</u> Good answers were either based on females, using $B(25, 0.45)$ and the hypotheses $H_0: p = 0.45$ and $H_1: p < 0.45$, or males, using $B(25, 0.55)$ and the hypotheses $H_0: p = 0.55$ and $H_1: p > 0.55$. Most candidates using a correct value of p then went on to compare 0.0639 ($P(X = 7)$ or $P(X = 18)$) with 0.05. Most then correctly stated that the result is not significant, or that H_0 should not be rejected, and then went on to give an appropriate conclusion in context. However, even good candidates often defined p incorrectly as the probability that a female (or male) driver was speeding, rather than the correct		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					definition as the probability that a driver speeding (or caught speeding) was female (or male). Partial credit was allowed for candidates who used the value of 0.5 for p. There were a considerable number of candidates who made little, or no, progress on this question. Exemplar 3 Let p = proportion of drivers at speed awareness course that are female $H_0: p = 0.45$ $H_1: p < 0.45$ $X \sim B(25, 0.45)$ $P(X \leq 9) = 0.06385$ As $0.06385 > 0.05$, we accept the H_0 There is not evidence to suggest male drivers are more likely to exceed the speed limit than female drivers. This candidate is set up and concluded the hypothesis test nicely. The conclusion is non-assertive and in context.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	d		eg Each driver speeding / caught speeding has the same (constant) probability of being selected for the speed awareness course.	E1 (AO2.4) [1]	OR the probability of any particular driver caught speeding being selected for the speed awareness course is independent of any other driver caught speeding being selected.		
					Examiner's Comments This part was often omitted or generic. The context of the question needed to be considered in the answer.		
					Exemplar 4 <u>Events occurred mutually and independently.</u> The candidate has offered up the idea of independence, but has not used this in the context of the question looking at the probability of any one driver caught speeding being selected is independent of any other driver caught speeding being selected.		
			Total	10			

Question		Answer/Indicative content	Marks	Part marks and guidance	
42	a	$\frac{1}{8} \times \frac{1}{7}$ $\frac{1}{56}$	M1 (AO1.1a) A1 (AO1.1) [2]	$o \frac{6!}{8!}$ r CAO accept 0.0179 0.0178571 42 Examiner's Comments Some candidates did this question 'with replacement', giving the answer to part (a) a $\frac{1}{64}$, and having incorrect s terms like $\left(\frac{5}{8}\right)^3$ in part (b). There were some good answers to part (b), usually relying on tree diagrams, though not all these candidates realised that they had to consider two cases: '3 women' and '2 women and 1 man'.	

Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	5C_3 or ${}^5C_2 \times {}^3C_1$ seen [10, 30] 8C_3 is denominator [56] {equiv to 8C_5 } $P(3, 0) + P(2, 1)$ attempted $\frac{5}{7}$ alternati $\frac{5}{8} \times \frac{4}{7} \times \frac{3}{6}$ or vely $3 \times \frac{5}{8} \times \frac{4}{7} \times \frac{3}{6}$ seen both correct including $\times 3$ $P(3, 0) + P(2, 1)$ attempted $\frac{5}{7}$	M1 (AO1.1a) B1 (AO1.1) M1 (AO2.1) A1 (AO1.1) [4] M1 (AO1.1a) A1 (AO1.1) M1 (AO2.1) A1 [4] (AO1.1)	OR 3W 10 ways 2W1M $10 \times 3 = 30$ ways 1W2M $5 \times 3 = 15$ ways 3M 1 way $S \frac{10+30}{10+30+15+1} = \frac{5}{7}$ o $\frac{60}{336}$ or $\frac{5}{28}$ or $\frac{180}{336}$ or $\frac{15}{28}$ Allow even if $\times 3$ missing	Allow M1 if the 2 cases are identified and added, whatever the calculation. Allow method by subtraction from 1. This question can be done by tree diagram, with similar working. Allow both M marks for all denoms $8 - \text{eg } \frac{5}{8} \times \frac{4}{8} \times \frac{3}{8} = \frac{60}{512}$ etc $\frac{15}{32}$ leading to	
		Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance																									
43	a		$0.6 + 0.5 - P(A \cap B) = 1 - 0.2$ oe soi $= 0.3$	M1 (AO1.1a) A1 (AO1.1) [2]	or M1 for probabilities in bold correct in table or marked correctly on Venn diagram NB 0.3 from 0.6×0.5 does not score	<table><tr><td></td><td colspan="3">A</td></tr><tr><td></td><td></td><td>A</td><td></td></tr><tr><td>B</td><td>0.3</td><td>0.2</td><td>0.5</td></tr><tr><td></td><td>0.3</td><td>0.2</td><td>0.5</td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td></td><td>0.6</td><td>0.4</td><td>1</td></tr></table>		A					A		B	0.3	0.2	0.5		0.3	0.2	0.5	B					0.6	0.4	1
	A																													
		A																												
B	0.3	0.2	0.5																											
	0.3	0.2	0.5																											
B																														
	0.6	0.4	1																											
	b		$\frac{0.3}{0.5}$ $= 0.6$	M1 (AO1.1) A1 (AO1.1) [2]	$\frac{\text{their } P(A \cap B)}{0.5}$																									

Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		independent since $p(A) = p(A/B)$ oe	B1 (AO2.4) [1]	or $0.6 \times 0.5 = 0.3$ or $P(A \cap B) = P(A) \times P(B)$ FT <i>their</i> values with correct argument	
			Examiner's Comments This question was generally very well done. Candidates who did less well were unable to explain why the events were independent in part (c). Candidates who did not do well assumed that A and B were independent in part (a) or worked with 0.2 instead of 0.8, going on to achieve a probability greater than 1 in part (b) – only a very small number of candidates recognised that this would indicate a mistake and went back to correct their earlier work.			
			Total	5		

Question		Answer/Indicative content	Marks	Part marks and guidance		
44	a	$2p + q + 0.2 + 0.3 = 1$ soi oe $2 \times p \times q = 0.06$ soi eliminate p or q with a correct substitution from one of <i>their</i> equations $q^2 - 0.5q + 0.06 = 0$ or $2p^2 - 0.5p + 0.03 = 0$ oe $q = 0.2$ or 0.3 and $p = 0.15$ or 0.1 $(q < 2p \text{ so}) q = 0.2$ and $p = 0.15$	B1 (AO2.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [6]	allow M1 if 2 omitted eg $2 \times \frac{0.03}{q} + q = 0.5$ or $2p + \frac{0.03}{p} = 0.5$ may be implied by eg $q = 0.2$ or 0.3 and $2p = 0.3$ or 0.2	NB if 2 omitted, A0 for $2p^2 - 0.5p + 0.6 = 0$ or $2q^2 + q + 0.24 = 0$ which have no real root	
	b	$10 \times q \times (1 - q)^9$ soi 0.27 or 0.268 or awrt 0.2684 isw	M1 (AO1.1) A1 (AO1.1) [2]	FT <i>their</i> q where $0 < q < 1$		

Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$H_0: p = 0.2$ $H_1: p > 0.2$ p is the probability that the spinner shows a 1 (on any given spin) oe use of $X \sim B(100, 0.2)$ where x is the number of 1s obtained in 100 spins to obtain $P(X \geq k)$ or $P(X \leq k)$ $P(X \leq 27) = \text{awrt } 0.97$ or $P(X \geq 28) = \text{awrt } 0.034$ $0.034 < 0.05$ or $0.97 > 0.95$ significant or reject H_0 or accept H_1; may be embedded in conclusion in context there is sufficient evidence to suggest (at 5% level) that the probability of a score of 1 is greater than 0.2	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) A1 (AO2.2b) [7]	both hypotheses ; allow equivalent in words or eg $P(1) = 0.2$ $k = 27, 28$ or 29 or critical region is $X \geq 28$ or 28 is in critical region must have the correct probability or correct critical region for the last two A marks	allow any parameter as long as clearly defined as probability M0 for $P(X = k)$ NB $P(X = 28) = 0.014...$ $P(X = 27) = 0.020168...$ FT <i>their</i> probability, dependent on award of first M1 do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; A0 if answer spoiled	
		Total	15			